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Artificial Intelligence in Depression-Medication Enhancement (AID-ME): A Cluster Randomized Trial of a Deep-Learning-Enabled Clinical Decision Support System for Personalized Depression Treatment Selection and Management.

D. Benrimoh, Kate Whitmore, Maud Richard, Grace Golden, Kelly Perlman, Sara Jalali, Timothy Friesen, Y. Barkat, J. Mehltretter, Robert Fratila, Caitrin Armstrong, Sonia Israel, Christina Popescu, Jordan F. Karp, Sagar V. Parikh, Shirin Golchi, Erica E. M. Moodie, Jun-Wei Shen, Anthony J. Gifuni, Manuela Ferrari, Mamta Sapra, S. Kloiber, Georges-F. Pinard, B. W. Dunlop, Karl Looper, Mohini Ranganathan, M. Enault, Serge Beaulieu, S. Rej, F. Hersson-Edery, Warren Steiner, A. Anacleto, S. Qassim, R. McGuire-Snieckus, H. Margolese

Aifred Health Inc, Montreal, Quebec, Canada.

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CLINICAL & AGENTIC AI TAKEAWAY

AI-enabled clinical decision support systems can potentially improve outcomes in moderate and greater severity major depressive disorder by personalizing treatment selection and management.

ABSTRACT

Background: There has been increasing interest in the use of artificial intelligence (AI)-enabled clinical decision support systems (CDSS) for the personalization of major depressive disorder (MDD) treatment selection and management, but clinical studies are lacking. We tested whether a CDSS that combines an AI which predicts remission probabilities for individual antidepressants and a clinical algorithm based on treatment can improve MDD outcomes. Methods: This was a multicenter, cluster randomized, patient-and-rater blinded and clinician-partially-blinded, active-controlled trial that recruited outpatient adults with moderate or greater severity MDD. All patients had access to a patient portal to complete questionnaires. Clinicians in the active group had access to the CDSS; clinicians in the active-control group received patient questionnaires; both groups received guideline training. Primary outcome was remission (<11 points on the Montgomery-Asberg Depression Rating Scale [MADRS]) at study exit. Results: Forty-seven clinicians were recruited at 9 sites. Of 74 eligible patients, 61 patients completed a postbaseline MADRS and were analyzed. There were no differences in baseline MADRS (P = .153). There were more remitters in the active (n = 12, 28.6%) than in the active-control (0%) group (P = .012, Fisher's exact). Of 3 serious adverse events, none were caused by the CDSS. Speed of improvement was higher in the active than the control group (1.26 vs 0.37, P = .03). Conclusions: While limited by sample size and the lack of primary care clinicians, these results demonstrate preliminary evidence that longitudinal use of an AI-CDSS can improve outcomes in moderate and greater severity MDD. Trial Registration: ClinicalTrials.gov identifier: NCT04655924.
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Study Overview & Background

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Methodology & Clinical Focus

Study Classification: rct. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

AI-enabled clinical decision support systems can potentially improve outcomes in moderate and greater severity major depressive disorder by personalizing treatment selection and management.

Extended Repository Abstract

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Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** D. Benrimoh, Kate Whitmore, Maud Richard, Grace Golden, Kelly Perlman, Sara Jalali, Timothy Friesen, Y. Barkat, J. Mehltretter, Robert Fratila, Caitrin Armstrong, Sonia Israel, Christina Popescu, Jordan F. Karp, Sagar V. Parikh, Shirin Golchi, Erica E. M. Moodie, Jun-Wei Shen, Anthony J. Gifuni, Manuela Ferrari, Mamta Sapra, S. Kloiber, Georges-F. Pinard, B. W. Dunlop, Karl Looper, Mohini Ranganathan, M. Enault, Serge Beaulieu, S. Rej, F. Hersson-Edery, Warren Steiner, A. Anacleto, S. Qassim, R. McGuire-Snieckus, H. Margolese
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